MODULE 25 · DOMAIN 3 - COST-OPTIMIZED ARCHITECTURE DESIGN

Pricing Models and Purchase Options

Match each slice of usage to the right purchase option: commitments for the baseline, Spot for the interruptible, reservations when capacity itself must be guaranteed.

DOMAIN 3 of 5 EXAM WEIGHT 15% MODULE 25 of 52

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A discount is not a capacity guarantee, and a capacity guarantee is not a discount. Know which one the question needs.

Purchase options at a glance

OPTION	BEST FOR	TRADE-OFF
On-Demand	Spiky, short-lived, unknown usage	Highest unit price
Savings Plans	Steady spend you can commit to	1 or 3 years of hourly spend
Reserved Instances	Steady use of a specific resource	Less flexible than Savings Plans
Spot	Interruptible, flexible work	Two-minute interruption notice
Capacity reservations	Capacity that must be available	Paid whether used or not

ON THE EXAM

Commit to the floor, run the peaks On-Demand, and push anything interruptible to Spot.

Savings Plans types

PLAN	COVERS	FLEXIBILITY
Compute	EC2, Fargate, Lambda	Any family, size, Region, OS
EC2 Instance	EC2 in one family and Region	Size and OS within the family
SageMaker AI	SageMaker AI instance usage	Any instance, Region, component
Database	Aurora, RDS, DynamoDB and six more	Engines, Regions, serverless or provisioned

ON THE EXAM

Usage that will change shape, family or Region: Compute Savings Plan. Locked to one family: EC2 Instance plan, bigger discount.

Database Savings Plans

- **Covers** Aurora, RDS, DynamoDB, ElastiCache, DocumentDB, Neptune, Keyspaces, Timestream, DMS.
- **One-year term**, committed hourly spend.
- **Up to 35%** on serverless, up to 20% on provisioned instances.
- **Flexible** across engines, Regions and deployment types.
- **Reserved DB instances** still exist, but lock to one instance class.

> ON THE EXAM
>
> Moving from provisioned to Aurora Serverless, or between engines: the Database Savings Plan keeps the discount.

Commitment discount or capacity guarantee?

SAVINGS PLANS AND REGIONAL RIS

Pay less

- Discount on matching usage
- No promise capacity exists
- Applies automatically by hour
- Unused commitment is still paid

ON-DEMAND CAPACITY RESERVATIONS

Be sure it launches

- Capacity held in one AZ
- Immediate or future-dated
- On-Demand price unless discounted
- Combines with Savings Plans

THE TIEBREAKER

Guaranteed launches for an event: a reservation. Then cover it with a Savings Plan for the discount.

Spot, used well

- **Up to 90% off** On-Demand, for spare capacity AWS can reclaim.
- **Two-minute notice**, plus an earlier rebalance recommendation.
- **Diversify** across many instance types and Availability Zones.
- **Price-capacity-optimised** allocation picks the deepest pools.
- **Fits** batch, CI, stateless web tiers, containers, checkpointed training.

> ON THE EXAM
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> Spot fails when a design asks for one instance type. Flexibility is what makes it reliable.

Capacity-based options

OPTION	USE IT FOR
On-Demand Capacity Reservation	Any duration, specific AZ, critical events or DR
Future-dated reservation	A known date ahead, such as a launch
Capacity Blocks for ML	GPU instances for a fixed future window
Provisioned throughput or capacity	DynamoDB, Kinesis, Bedrock: pay for a rate

ON THE EXAM

GPUs for a training run starting on a known date, for a fixed period: Capacity Blocks for ML.

Sizing and managing commitments

- **Recommendations** in Cost Explorer from past usage, by plan type and term.
- **Coverage** shows how much usage is committed; **utilisation** how much commitment is used.
- **Commit to the floor**, not the average; buy more in steps.
- **Return a Savings Plan** within 7 days, same month, up to 100 dollars an hour.
- **Savings Plans cannot be resold**; Standard EC2 RIs can, on the Marketplace.

> ON THE EXAM
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> Low utilisation means over-commitment. Low coverage means money left on the table.

Reading the scenario

WHEN THE QUESTION SAYS	IT IS TESTING
Moving from EC2 to Fargate or Lambda	Compute Savings Plan
Fault-tolerant, flexible batch	Spot, diversified
Must launch on a specific day, in one AZ	Capacity reservation
Databases changing engine or going serverless	Database Savings Plan
GPUs for a fixed window, future date	Capacity Blocks for ML
Bought the wrong plan yesterday	Return within 7 days

ON THE EXAM

Read the usage pattern first: steady, spiky, interruptible, changing, or capacity-critical.

> Meridian Freight spends a steady 40 dollars an hour on EC2 across three Regions, plus Fargate and Lambda. Over the next year it plans to move most EC2 workloads to Fargate.
>
> Finance wants the largest discount that will still apply after the migration.

Which purchase option should the solutions architect recommend?

A. EC2 Instance Savings Plans for each family and Region

B. Standard Reserved Instances for the current instance types

C. Spot Instances for all EC2 workloads

D. A Compute Savings Plan sized to the steady hourly spend

◆ PAUSE AND DECIDE

Which purchase option should the solutions architect recommend?

A EC2 Instance plans

Deeper discount, but stranded when workloads move to Fargate.

B Standard RIs

Locked to instance types that are being retired.

C Spot

Not suitable for all workloads, and not a commitment.

D Compute Savings Plan

Applies to EC2, Fargate and Lambda in any Region.

THE SIGNAL

Usage changing from EC2 to containers or functions: Compute Savings Plan.

> Meridian Freight renders route-optimisation maps overnight. Jobs checkpoint every few minutes, can restart on any instance, and run on many compute-optimised instance types.
>
> The team wants the lowest possible compute cost.

What should the solutions architect recommend?

A. Spot Instances in an Auto Scaling group spanning many instance types and AZs, with price-capacity-optimised allocation

B. On-Demand Instances of one instance type

C. Three-year Standard Reserved Instances

D. On-Demand Capacity Reservations for the overnight window

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Diversified Spot

Deepest discount; checkpoints and diversity absorb interruptions.

B On-Demand, one type

Full price for work that tolerates interruption.

C 3-year RIs

Pays for 24 hours a day for an overnight job.

D Capacity Reservations

Buys certainty the workload does not need, at full price.

THE SIGNAL

Checkpointed, restartable, flexible on instance type: diversified Spot.

> Meridian Freight holds a Compute Savings Plan. Next month, a three-day peak-season sale will need 200 extra instances of a specific type in two Availability Zones.
>
> Leadership requires certainty that the instances will launch on those days.

What should the solutions architect do?

A Rely on the Compute Savings Plan to secure the capacity

B Buy one-year zonal Reserved Instances for the 200 instances

C Create future-dated On-Demand Capacity Reservations for the three days in both AZs

D Request Spot capacity for the sale

◆ PAUSE AND DECIDE

What should the solutions architect do?

A Savings Plan

Discounts usage; does not hold capacity.

B 1-year zonal RIs

Hold capacity, but pay for a year to cover three days.

C Future-dated reservations

Capacity held for exactly the days needed; the plan still discounts it.

D Spot

The opposite of certainty.

THE SIGNAL

Certainty for a known window: a future-dated reservation. The Savings Plan still discounts it.

> Meridian Freight runs Aurora PostgreSQL provisioned clusters and DynamoDB in two Regions. Over the next year it will move several clusters to Aurora Serverless v2 and may add a third Region.
>
> Finance wants a commitment discount that keeps applying through these changes.

Which purchase option should the solutions architect recommend?

A Reserved DB instances for the current Aurora instance classes

B A Database Savings Plan sized to the steady database spend

C A Compute Savings Plan

D DynamoDB reserved capacity only

◆ PAUSE AND DECIDE

Which purchase option should the solutions architect recommend?

A Reserved DB instances

Tied to instance classes that become serverless.

B Database Savings Plan

Follows spend across Aurora, DynamoDB, serverless and Regions.

C Compute Savings Plan

Does not cover database services.

D DynamoDB reserved only

Leaves Aurora uncovered.

THE SIGNAL

Databases changing shape or Region: the Database Savings Plan.

> Meridian Freight's data science team must fine-tune a forecasting model on 16 GPU instances for two weeks, starting on the first of next month.
>
> The run cannot be interrupted, and the team does not need the GPUs afterwards.

What should the solutions architect recommend?

A. Spot GPU instances with checkpointing

B. A one-year EC2 Instance Savings Plan for the GPU family

C. EC2 Capacity Blocks for ML for the two-week window

D. On-Demand Instances launched on the day

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Spot GPUs

The run must not be interrupted.

B 1-year Savings Plan

Discounts a year of spend for two weeks of use, and holds no capacity.

C Capacity Blocks for ML

Reserved GPUs for exactly the window, from a future date.

D On-Demand on the day

No guarantee scarce GPUs will be available.

THE SIGNAL

Uninterrupted GPUs, fixed window, known start date: Capacity Blocks for ML.

> Yesterday, a Meridian Freight engineer bought a one-year EC2 Instance Savings Plan with a commitment of 30 dollars an hour, for the wrong instance family.
>
> Finance asks what can be done.

What should the solutions architect advise?

A. Return the Savings Plan, since it is within seven days, in the same month, and under 100 dollars an hour

B. Sell it on the Reserved Instance Marketplace

C. Exchange it for a Compute Savings Plan

D. Nothing; Savings Plans can never be cancelled

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A Return it

Meets every condition: 7 days, same month, 100 dollars an hour or less.

B Marketplace

Only Standard EC2 RIs can be sold; Savings Plans cannot.

C Exchange

Exchanges apply to Convertible RIs, not Savings Plans.

D Nothing

Out of date since returns were introduced in 2024.

THE SIGNAL

A recent, small Savings Plan mistake can be returned. Older or larger ones cannot.

What to carry out of this module

- **Discounts and capacity are separate**; reservations combine with Savings Plans.
- **Compute Savings Plans** for changing compute; EC2 Instance plans for fixed families.
- **Database Savings Plans** follow databases across engines, Regions, serverless.
- **Spot** for interruptible work, diversified across types and AZs.
- **Capacity Blocks** for GPU windows; return mistaken plans within 7 days.

> ON THE EXAM
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> Name the usage pattern, then match the option. Most estates layer several.